


**Sharif University of Technology**
**Department of Electrical Engineering**

# Introduction to Machine Learning

## Fall 2025 Semester

## Project Phase 1

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# 1 Question 1

Assume the in-distribution (ID) data are modeled by a single-component Gaussian model

$$p_{ID}(x) = \mathcal{N}(x; \mu, \Sigma), \qquad x \in \mathbb{R}^d,$$

with mean $\mu \in \mathbb{R}^d$ and covariance $\Sigma \in \mathbb{R}^{d \times d}$, where $\Sigma \succ 0$.

## 1.1 Write $\log p_{ID}(x)$ explicitly

The multivariate Gaussian density is

$$p_{ID}(x) = \frac{1}{(2\pi)^{d/2}\,|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right).$$

Taking $\log(\cdot)$ yields

$$\boxed{\log p_{ID}(x) = -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma| - \frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)}.$$

## 1.2 Show $\log p_{ID}(x) \geq \tau$ is an ellipsoid condition

Start from the thresholding rule

$$\log p_{ID}(x) \geq \tau.$$

Substitute the explicit expression above:

$$-\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma| - \frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu) \geq \tau.$$

Move the constant terms to the right-hand side:

$$-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu) \geq \tau + \frac{d}{2}\log(2\pi) + \frac{1}{2}\log|\Sigma|.$$

Multiply both sides by $-2$ (note the inequality flips):

$$(x-\mu)^\top \Sigma^{-1}(x-\mu) \leq -2\tau - d\log(2\pi) - \log|\Sigma|.$$

The left-hand side is the squared Mahalanobis distance

$$d_M^2(x, \mu; \Sigma) \;=\; (x-\mu)^\top \Sigma^{-1}(x-\mu).$$

For $\Sigma \succ 0$, the matrix $\Sigma^{-1} \succ 0$ defines a positive-definite quadratic form, so the set

$$\{x : \; (x-\mu)^\top \Sigma^{-1}(x-\mu) \leq C\}$$

is a (hyper-)ellipsoid centered at $\mu$, with principal axes aligned with eigenvectors of $\Sigma$ and axis lengths proportional to $\sqrt{C\,\lambda_j(\Sigma)}$.

### 1.3 Exact equation of the ellipsoid in terms of $\mu$, $\Sigma$, and $\tau$

Define

$$C(\tau) \;=\; -2\tau - d\log(2\pi) - \log|\Sigma|.$$

Then the boundary corresponding to $\log p_{ID}(x) = \tau$ is

$$(x-\mu)^\top \Sigma^{-1}(x-\mu) = C(\tau),$$

and the acceptance region $\log p_{ID}(x) \geq \tau$ is the interior plus boundary:

$$(x-\mu)^\top \Sigma^{-1}(x-\mu) \leq C(\tau).$$

$$\boxed{\log p_{ID}(x) \geq \tau \quad \Longleftrightarrow \quad (x-\mu)^\top \Sigma^{-1}(x-\mu) \leq -2\tau - d\log(2\pi) - \log|\Sigma|}$$

$$\boxed{\text{Ellipsoid boundary } (\log p_{ID}(x) = \tau): \qquad (x-\mu)^\top \Sigma^{-1}(x-\mu) = -2\tau - d\log(2\pi) - \log|\Sigma|}$$

$$\boxed{\text{Ellipsoid (inside/on):} \qquad \{x \in \mathbb{R}^d : \; (x-\mu)^\top \Sigma^{-1}(x-\mu) \leq -2\tau - d\log(2\pi) - \log|\Sigma|\}}$$

## 2 Question 2

Consider a Gaussian Mixture Model (GMM) with $K$ components and parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, where $\pi_k \geq 0$ and $\sum_{k=1}^K \pi_k = 1$. For a given sample $x \in \mathbb{R}^d$, the (marginal) likelihood under the mixture is

$$p(x) = \sum_{k=1}^K \pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k), \qquad \ell(x) \;=\; \log p(x).$$

### 2.1 Why $\log\Big(\sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k)\Big)$ is not a sum of component log-likelihoods

A common (incorrect) attempt is to "distribute" the logarithm over the mixture sum:

$$\log\left(\sum_{k=1}^K \pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k)\right) \stackrel{?}{=} \sum_{k=1}^K \log(\pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k)) \,.$$

This is not valid because the logarithm is nonlinear and, in general, for positive numbers $\{a_k\}$,

$$\log\left(\sum_{k=1}^K a_k\right) \neq \sum_{k=1}^K \log a_k.$$

In fact, equality would require very special situations (e.g., effectively only one nonzero term). In a mixture model the component contributions $\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)$ are added *before* applying log, so the parameters of all components appear coupled inside the same $\log(\cdot)$ expression. This "log-of-sum" structure is exactly what prevents direct closed-form maximum likelihood updates for $\{\pi_k, \mu_k, \Sigma_k\}$.

$$\boxed{\log\left(\sum_{k=1}^K \pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k)\right) \neq \sum_{k=1}^K \log(\pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k)) \quad \text{(in general).}}$$

## 2.2 Jensen lower bound on the log-likelihood

Introduce an auxiliary categorical distribution $q(z)$ over the latent component index $z \in \{1, \dots, K\}$, with $q(k) \geq 0$ and $\sum_{k=1}^{K} q(k) = 1$. Rewrite the mixture likelihood using $q$:

$$p(x) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k) = \sum_{k=1}^{K} q(k) \, \frac{\pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k)}{q(k)} = \mathbb{E}_q \left[ \frac{\pi_z \, \mathcal{N}(x; \mu_z, \Sigma_z)}{q(z)} \right].$$

Since $\log(\cdot)$ is concave, Jensen's inequality gives

$$\log p(x) = \log \mathbb{E}_q \left[ \frac{\pi_z \, \mathcal{N}(x; \mu_z, \Sigma_z)}{q(z)} \right] \geq \mathbb{E}_q \left[ \log \frac{\pi_z \, \mathcal{N}(x; \mu_z, \Sigma_z)}{q(z)} \right].$$

Writing the expectation explicitly,

$$\log p(x) \geq \sum_{k=1}^{K} q(k) \left( \log \pi_k + \log \mathcal{N}(x; \mu_k, \Sigma_k) - \log q(k) \right).$$

The right-hand side is the standard evidence lower bound (ELBO) for a single sample:

$$\boxed{\log \left( \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x; \mu_k, \Sigma_k) \right) \geq \sum_{k=1}^{K} q(k) \Big( \log \pi_k + \log \mathcal{N}(x; \mu_k, \Sigma_k) - \log q(k) \Big).}$$

Equivalently, separating terms,

$$\sum_{k=1}^{K} q(k) \big( \log \pi_k + \log \mathcal{N}(x; \mu_k, \Sigma_k) \big) \; + \; H(q), \qquad H(q) = -\sum_{k=1}^{K} q(k) \log q(k),$$

so the bound trades off a weighted expected log component-likelihood and the entropy of $q$.

## 2.3 Connection to the EM objective

Let $\theta = \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^{K}$. The inequality above can be written as

$$\log p(x; \theta) \; \geq \; \mathcal{L}(q, \theta),$$

where $\mathcal{L}(q, \theta)$ denotes the lower bound (ELBO). A useful identity refines this relationship:

$$\log p(x; \theta) = \mathcal{L}(q, \theta) \; + \; \mathrm{KL}\big( q(z) \,\|\, p(z \mid x; \theta) \big),$$

and since $\mathrm{KL}(\cdot \| \cdot) \geq 0$, $\mathcal{L}(q, \theta)$ is indeed a lower bound.

The EM algorithm can be viewed as coordinate ascent on $\mathcal{L}(q, \theta)$:

$$\boxed{\text{E-step: } q^{(t)}(z) \leftarrow p(z \mid x; \theta^{(t)}), \qquad \text{M-step: } \theta^{(t+1)} \leftarrow \arg\max_{\theta} \mathcal{L}(q^{(t)}, \theta).}$$

In the E-step, choosing $q^{(t)}(z) = p(z \mid x; \theta^{(t)})$ makes the KL term zero, hence the bound becomes tight at $\theta^{(t)}$. In the M-step, maximizing $\mathcal{L}(q^{(t)}, \theta)$ increases (or keeps) the bound, and because $\log p(x; \theta) = \mathcal{L}(q^{(t)}, \theta) + \mathrm{KL}(\cdot) \geq \mathcal{L}(q^{(t)}, \theta)$ with equality at the previous iterate, the true log-likelihood cannot decrease across EM iterations.

EM maximizes the same lower bound obtained via Jensen's inequality.
The E-step makes this bound tight by matching the posterior responsibilities,
and the M-step increases the bound by optimizing the model parameters.

# 3 Question 3

During the M-step of EM, the parameters are updated by maximizing the expected complete-data log-likelihood. Let $X = \{x_i\}_{i=1}^n$ with $x_i \in \mathbb{R}^d$, and let $z_i \in \{1, \dots, K\}$ be the latent component index. At iteration $t$, define the responsibilities

$$\gamma_{ik}^{(t)} \;=\; p\big(z_i = k \mid x_i; \theta^{(t)}\big)\,, \qquad N_k^{(t)} \;=\; \sum_{i=1}^{n} \gamma_{ik}^{(t)}.$$

## 3.1 Obtaining the mean update by maximizing the expected complete-data log-likelihood

The complete-data log-likelihood can be written using indicator variables $z_{ik} = \mathbb{I}\{z_i = k\}$:

$$\log p(X, Z; \theta) = \sum_{i=1}^{n} \sum_{k=1}^{K} z_{ik} \Big( \log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k) \Big).$$

In EM we maximize its expectation under the posterior of $Z$ given $X$ and the current parameters $\theta^{(t)}$:

$$Q(\theta, \theta^{(t)}) \;=\; \mathbb{E}_{Z|X,\theta^{(t)}} \big[ \log p(X, Z; \theta) \big] = \sum_{i=1}^{n} \sum_{k=1}^{K} \gamma_{ik}^{(t)} \Big( \log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k) \Big).$$

To update $\mu_k$, all terms not depending on $\mu_k$ can be treated as constants, so the relevant part is

$$Q_k(\mu_k) = \sum_{i=1}^{n} \gamma_{ik}^{(t)} \log \mathcal{N}(x_i; \mu_k, \Sigma_k).$$

Using the Gaussian log-density,

$$\log \mathcal{N}(x_i; \mu_k, \Sigma_k) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k),$$

and dropping constants with respect to $\mu_k$, maximizing $Q_k(\mu_k)$ is equivalent to minimizing

$$\sum_{i=1}^{n} \gamma_{ik}^{(t)} (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k).$$

This shows the mean update comes from optimizing the expected complete-data objective for the $k$-th component.

## 3.2 Gradient w.r.t. $\mu_k$ and setting it to zero

Define

$$J(\mu_k) = \sum_{i=1}^{n} \gamma_{ik}^{(t)} (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k).$$

Since $\Sigma_k^{-1}$ is symmetric, the gradient of one term is

$$\nabla_{\mu_k} \Big( (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k) \Big) = -2\, \Sigma_k^{-1} (x_i - \mu_k).$$

Therefore,

$$\nabla_{\mu_k} J(\mu_k) = \sum_{i=1}^{n} \gamma_{ik}^{(t)} \big( -2\Sigma_k^{-1}(x_i - \mu_k) \big) = -2\Sigma_k^{-1} \sum_{i=1}^{n} \gamma_{ik}^{(t)} (x_i - \mu_k).$$

Setting the gradient to zero and using the invertibility of $\Sigma_k$,

$$\sum_{i=1}^{n} \gamma_{ik}^{(t)} (x_i - \mu_k) = 0 \implies \sum_{i=1}^{n} \gamma_{ik}^{(t)} x_i = \mu_k \sum_{i=1}^{n} \gamma_{ik}^{(t)} = \mu_k \, N_k^{(t)}.$$

Hence,

$$\boxed{\mu_k^{(t+1)} = \frac{1}{N_k^{(t)}} \sum_{i=1}^{n} \gamma_{ik}^{(t)} x_i}.$$

### 3.3 Why the update is a weighted average

The coefficients

$$w_i = \frac{\gamma_{ik}^{(t)}}{N_k^{(t)}}$$

satisfy $w_i \geq 0$ and $\sum_{i=1}^{n} w_i = 1$. Therefore,

$$\mu_k^{(t+1)} = \sum_{i=1}^{n} w_i x_i$$

is a convex combination of the data points, i.e., the center of mass of $\{x_i\}$ where each point is weighted by its responsibility for component $k$.

$$\boxed{\mu_k^{(t+1)} \text{ is a weighted average of } \{x_i\} \text{ with weights } w_i = \gamma_{ik}^{(t)} / N_k^{(t)}.}$$

## 4 Question 4

Assume the in-distribution (ID) data are drawn from a $d$-dimensional Gaussian distribution

$$x \sim \mathcal{N}(\mu, \Sigma), \qquad \Sigma = \sigma^2 I.$$

### 4.1 Expected log-likelihood and its scaling with $d$

For the isotropic Gaussian, the log-likelihood of a single sample $x$ is

$$\log p(x) = -\frac{d}{2} \log(2\pi) - \frac{d}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \|x - \mu\|^2.$$

Taking expectation with respect to $x \sim \mathcal{N}(\mu, \sigma^2 I)$, the first two terms are constants. For the quadratic term,

$$\|x - \mu\|^2 = \sum_{j=1}^{d} (x_j - \mu_j)^2, \qquad \mathbb{E}\big[(x_j - \mu_j)^2\big] = \sigma^2,$$

hence

$$\mathbb{E}\big[\|x - \mu\|^2\big] = d\sigma^2.$$

Substituting into the expectation of the log-likelihood,

$$\mathbb{E}[\log p(x)] = -\frac{d}{2}\log(2\pi) - \frac{d}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(d\sigma^2) = -\frac{d}{2}\log(2\pi\sigma^2) - \frac{d}{2}.$$

$$\boxed{\mathbb{E}[\log p(x)] = -\frac{d}{2}\log(2\pi\sigma^2) - \frac{d}{2},}$$

which shows that the expected log-likelihood decreases linearly with the dimension $d$.

## 4.2 Why an OOD sample can have higher likelihood than an ID sample

Consider an OOD sample $x_{\text{OOD}}$ that has a large norm but exhibits strong regularity. Suppose that in most coordinates $x_{\text{OOD}}$ is extremely close to $\mu$, while deviation occurs mainly along a small number of directions. Decompose the squared distance as

$$\|x_{\text{OOD}} - \mu\|^2 = \varepsilon_{\text{small}} + R^2_{\text{large}},$$

where $\varepsilon_{\text{small}}$ aggregates many near-zero deviations and $R^2_{\text{large}}$ corresponds to a few large but structured deviations.

For a typical ID sample $x_{\text{ID}}$, the squared norm satisfies

$$\|x_{\text{ID}} - \mu\|^2 \sim \sigma^2\chi^2_d, \qquad \mathbb{E}\big[\|x_{\text{ID}} - \mu\|^2\big] = d\sigma^2,$$

and in high dimensions it concentrates sharply around this value. The log-likelihoods can be compared via

$$\log p(x) = -\frac{d}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|x - \mu\|^2.$$

Therefore,

$$\log p_{ID}(x_{\text{OOD}}) > \log p_{ID}(x_{\text{ID}}) \quad \Longleftrightarrow \quad \|x_{\text{OOD}} - \mu\|^2 < \|x_{\text{ID}} - \mu\|^2.$$

Even if $x_{\text{OOD}}$ is far from $\mu$ in a few coordinates, its total squared distance can be smaller than that of a typical ID sample if it is unusually close to $\mu$ in the remaining $d-1$ (or more) dimensions. This situation becomes increasingly plausible as $d$ grows.

> In high dimensions, a structured OOD sample can satisfy
> $$\|x_{\text{OOD}} - \mu\|^2 < \|x_{\text{ID}} - \mu\|^2,$$
> and therefore receive a higher likelihood under the ID model.

## 4.3 Why likelihood alone is unreliable for OOD detection in high dimensions

The Gaussian density depends on the squared Euclidean distance through

$$p(x) \propto \exp\left(-\frac{1}{2\sigma^2}\|x - \mu\|^2\right).$$

In $d$ dimensions, most ID probability mass lies in a thin shell of radius approximately $\sqrt{d}\sigma$ around $\mu$ rather than near the mode. As $d$ increases, the density at this typical radius decreases exponentially, even though these points are fully in-distribution.

At the same time, OOD samples with low intrinsic dimensionality or strong regularity may lie unusually close to $\mu$ in most coordinates, leading to a smaller total squared distance and thus a higher likelihood than that of typical ID samples.

Because likelihood depends on the sum of squared deviations across all dimensions, high-dimensional concentration of measure allows structured OOD samples to achieve likelihoods comparable to or even higher than typical ID samples.
Hence, likelihood alone is not a reliable criterion for OOD detection in high-dimensional spaces.